

Instrument Control and Cycle Test Scripts

This repository contains two Python scripts for automating hardware test setups:

- `check_port.py` — Scans all available serial (COM) and VISA (USB/GPIB/etc.) instruments, identifies them via `*IDN?`, and prints a summary.
- `cycle_test.py` — Runs an automated charge/discharge cycle test on a UPS (Uninterruptible Power Supply) while logging measurements from multiple instruments to a CSV file.

Requirement

- Python
 - `pyserial`
 - `pyvisa`
- NI VISA Driver (To identify VISA port)

How to run

1. Open Terminal by clicking Windows button and X button, then I button.
2. Navigate to the directory when the python script exists.
 - a. Use `'ls'` or `'ll'` to look for file and folder in the current directory
 - b. Use `'cd'` to navigate to different path.
3. Run `'python check_port.py'` to check all the port detected. All the port with device information will be displayed.
4. Edit the `cycle_test.py` with Notepad, change to port to match the respective device to model, and then save it.
5. Run `'python cycle_test.py'`.

Potential Issues & Solution

- Some device might not present even it is physically connected.
 - Restart the device should solve the problem
- Device with VISA connection not present.
 - Install the NI VISA Driver